

Implicit associations between pain and self-schema in patients with chronic pain



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ABSTRACT

Chronic pain often interferes with daily functioning, and may become a threat to an individual's sense of self. Despite the development of a recent theoretical account focussing upon the relationship between the presence of chronic pain and a person's self, research investigating this idea is limited. In the present study we aimed to (1) compare the strength of association between self- and pain schema in patients with chronic pain and healthy control subjects and (2) research whether the strength of association between self- and pain-schema is related to particular pain-related outcomes and individual differences of patients with chronic pain. Seventy-three patients with chronic pain ($M_{\text{age}} = 49.95$; $SD = 9.76$) and 53 healthy volunteers ($M_{\text{age}} = 48.53$; $SD = 10.37$) performed an Implicit Association Test (IAT) to assess the strength of association between pain- and self-schema. Patients with chronic pain also filled out self-report measures of pain severity, pain suffering, disability, depression, anxiety, acceptance, and helplessness. Results indicated that the pain- and self-schema were more strongly associated in patients with chronic pain than in healthy control subjects. Second, results indicated that, in patients with chronic pain, a stronger association between self- and pain-schema, as measured with the IAT, is related to a heightened level of pain severity, pain suffering, anxiety, and helplessness. Current findings give first support for the use of an IAT to investigate the strength of association between self- and pain-schema in patients with chronic pain and suggest that pain therapies may incorporate techniques that intervene on the level of self-pain enmeshment.

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1. Introduction

Chronic pain often interferes with daily life activities [17,24,45]. Lasting pain may also influence the individual's sense of self (i.e., an individual's self-schema) [5,24,33]. Indeed, the fact that pain persists and remains to interrupt and interfere with daily functioning may be damaging to one's sense of self, and result in suffering [23]. A theoretical account in this context is the schema enmeshment model of pain [33]. A central tenet of this model relates to strength of association between a person's pain- and self-schema (i.e., mental structures constructed through experience used to process incoming stimuli). It is assumed that the repeated and simultaneous activation of the content of the self- and pain-schema, as is the case in patients with chronic pain, results in a stronger association between a person's pain- and self-schema. Furthermore, it is proposed that such a strong association is

detrimental for pain outcomes (i.e., disability, pain suffering) [21,25,33] or related to chronic pain patient characteristics (e.g., depressive mood, anxiety, acceptance) [25,33,34,37,42]. Research investigating the above-mentioned topic is, however, still in its infancy [25].

As yet, the association between pain- and self-schema in patients with chronic pain has mainly been investigated by means of explicit (e.g., interview) and semi-explicit measures (e.g., Sentence Completion Test) [42,37,38]. Although these studies provide some evidence for the idea that chronic pain influences the individual's sense of self, there are limitations to the use of (semi-)explicit measures. Indeed, these measures tap only conscious cognitive processes, and explicit measures may be more vulnerable for response bias. Researchers have therefore developed so-called implicit measures that are less susceptible to bias and can reveal associations between schemata even when people are unwilling or unable to report those associations [11,30].

The main aim of the current research was to investigate the strength of association between pain- and self-schema in patients

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with chronic pain and healthy control subjects. To assess the strength between self- and pain-schema, we used the Implicit Association Test (IAT) [14]. The basic idea is that people are faster to categorize stimuli related to two associated concepts in the same way (e.g., by pushing the same button) than to categorize these stimuli in a different way (e.g., by pushing a different button) [14]. In this study, participants were required to categorize words related to the self - other dimension (e.g., <participant's first name> - <unfamiliar first name>), and words related to the pain - free of pain dimension (e.g., "excruciating" - "relieving").

We hypothesized that the association between pain- and self-schema, as measured by the IAT, is stronger in patients with chronic pain than in healthy control subjects. Furthermore, we also hypothesized that within the group of patients with chronic pain a stronger association between pain- and self-schema would be related to worse pain-related outcomes, in particular disability and suffering from pain. Finally, we tested whether IAT scores are related to depressive mood and level of anxiety and acceptance of the patients with chronic pain and so replicate previous research using (semi-)explicit measures to assess the overlap between pain- and self-schema in patients with chronic pain [25,42].

2. Method

2.1. Participants

Patients with chronic pain were recruited via an invitation letter sent to the members of the Flemish Pain League. Five-hundred and eighteen members responded to the letter, of which 315 agreed to be contacted by phone. In the period February-March 2011, 267 persons were actually contacted by telephone. Inclusion criteria for patients with chronic pain were: (1) aged between 18 and 65 years; (2) sufficient knowledge of the Dutch language; and (3) suffering from pain that lasted for at least 6 months. Individuals were excluded when headache was the most important pain (cfr. [12]), when they were unable to use both index fingers, or when their eyesight was not normal or corrected-to-normal (e.g., by glasses) [46]. Eighty-one patients with chronic pain who fulfilled the criteria agreed to participate. Because participants needed to travel to the university campus to participate in this study, transportation problems were mentioned as the most frequent reason for non-participation. However, later on, a further 7 patients decided not to participate because of health problems, and 1 participant could not execute the IAT because of insufficient time to complete the task during the experiment session. The final chronic pain sample consisted of 73 individuals. A control group matched for age and gender (on group level) was recruited via advertisement in a local newspaper and via flyers. A total of 86 individuals contacted the researcher to participate in the study. Inclusion and exclusion criteria were similar to those in the chronic pain group, except for (1) age range which was between 21 and 65 years (because of matching with patients with chronic pain) and (2) participants should not report current pain problems. A total of 54 participants were eligible to participate in the study. The main reasons for exclusion were age range ($n = 13$) and presence of a current pain problem ($n = 12$). The final healthy control sample consisted of 53 individuals.

Both groups were recruited as part of the Ghent Pain and Disability Study I (GPD-I-study). A flowchart and more details of the recruitment and procedure of the GPD-I-study are available at <http://hdl.handle.net/1854/LU-3050986>. The study design was approved by the ethics committee of the Faculty of Psychology and Educational Sciences of Ghent University, and written informed consent was obtained from all participants. All participants received a monetary reward for participation.

2.2. Questionnaires

Disability was assessed by means of the Dutch version of the Pain Disability Index (PDI; [35]). In this questionnaire, participants are asked to indicate the extent of disability experienced in 7 areas of everyday life (e.g., family/home responsibilities and social activity) using 0–10 Likert scales (0 = no disability and 10 = total disability). Scores range from 0 to 70. The reliability and validity of the PDI have been well established [43]. In the present study, Cronbach's alpha of the PDI was 0.81.

Depressive and anxious mood were measured using the Hospital Anxiety and Depression Scale (HADS; [48,49]). The HADS is a self-report scale that screens for the presence of depression and anxiety during the past week. The HADS was especially designed to measure depression and anxiety among patients with "medical conditions" [48]. The HADS-D (depression subscale) consists of 7 items that are rated on a 4-point Likert scale (e.g., I feel cheerful). Scores vary between 0 and 21. Also the HADS-A (anxiety subscale) consists of 7 items that are rated on a 4-point Likert scale (e.g., I feel tense or wound up). Again scores vary between 0 and 21. The HADS was found to perform well in assessing the symptom severity of depression and anxiety in somatic and primary care patients [3]. Cronbach's alpha of the HADS-D and HADS-A in the present study were 0.82 and 0.80 respectively.

Participants' level of anxiety was further assessed via the Dutch version of the State-Trait Anxiety Inventory (STAI; [40,47]) because the STAI measures anxiety in a more elaborated way than the HADS-A. The STAI measures State anxiety as well as Trait anxiety. The STAI-trait (STAI-T) subscale measures the disposition toward anxiety as a personality trait, which is defined as the relatively stable individual difference in anxiety proneness. The STAI-state (STAI-S) subscale measures the intensity of anxiety as a current emotional state consisting of subjective feelings of tension, nervousness, apprehension, and worry, and activation or arousal of the autonomic nervous system. The STAI consists of 40 items in which people are asked to report their feelings in general (e.g., I feel happy) and at present (e.g., I feel upset) using a 4-point Likert scale. Scores for the state and the trait version vary between 20 and 80. This questionnaire consistently demonstrated adequate psychometric properties and is among the most commonly used measures of anxiety [2,9,26,27,41]. In the present study, Cronbach's alpha of the STAI-S (STAI state version) and STAI-T (STAI trait version) were 0.94 and 0.94 respectively.

Pain severity and Pain suffering were assessed using the Multidimensional Pain Inventory (MPI; [18,19]). Part I of the MPI consists of 5 subscales assessing the impact of pain (i.e., pain severity, pain interference, social support, perceived life control, and affective distress). Pain severity was assessed by means of 2 items (i.e., "Rate the level of your pain at the present moment" and "On average, how severe has your pain been during the last week."). We opted to use only 2 items of the MPI severity subscale because the third item (i.e., How much suffering do you experience because of your pain?) relates to suffering rather than pain severity (see [32]). This item was coded and reported as pain suffering. The reliability and validity of the MPI have been well established [36]. In the present study Cronbach's alpha of the MPI pain severity subscale was 0.95.

Helplessness (i.e., the tendency to focus on the adverse aspects of the disease/pain and to generalize them to daily functioning) and Acceptance (i.e., the tendency to recognize the need to adapt to a chronic disease/pain while perceiving the ability to tolerate and manage its adverse consequences) were assessed using the 6-item Helplessness subscale and the 6-item Acceptance subscale of the Illness Cognitions Questionnaire (ICQ; [10]) respectively. The ICQ showed a good reliability and validity [20]. In the

present study, Cronbach's alpha of the Helplessness subscale and the Acceptance subscale of the ICQ were 0.85 and 0.90 respectively.

2.3. Word stimuli

The “self” category was represented by three words characterising the participant (his/her first name, his/her surname, and his/her place of residence). The “other” category was represented by the first name, surname, and place of residence of a completely unknown and fictive person chosen by participants from a set of possible stimuli. The “pain” and “free of pain” categories were each represented by 3 adjectives representing these dimensions (pain: excruciating [kwellend], horrible [vreselijk], exhausting [uitputtend]; free of pain: relaxing [ontspannend], relieving [bevrijdend], lenitive [verzachtend]). Pain-related adjectives were derived from the McGill Pain Questionnaire [22]. The adjectives belonging to the “free of pain” category were derived from previous IAT research [16] (Fig. 1).

2.4. Implicit Association Test

In line with Greenwald and colleagues [15], the IAT consisted of 7 blocks, of which the third and fourth as well as the sixth and seventh were critical. Each block started with the presentation of the relevant category-labels for 3000 ms. Labels were “me [ik]” and “not-me [niet-ik]” for the self-other dimension, and “pain [pijn]” and “free of pain [pijnvrij]” for the pain - free of pain dimension. In each block, the relevant labels were shown in the upper left and right corners of the screen. The labels remained on the screen for the entire duration of each block (see Fig. 1). In the first block, participants practised the discrimination between self and other stimuli. Each “self”- and “other”- related word was presented 4 times (i.e., 24 trials). In the second block, participants practised the discrimination between “pain” and “free of pain” stimuli. Each “pain”- and “free of pain”-related word was presented 4 times (i.e., 24 trials). The third and fourth blocks were the self-pain blocks. In these blocks, participants categorised as quickly and as accurately as possible the words of all four categories. Pain words were categorized by pressing the same key that was used for “self”-related

words, whereas “free of pain” words were categorized by pressing the same key that was used for “other”-related words. Each word was presented 4 times in each of the self-pain blocks (i.e., 48 trials/block). In the fifth block, participants needed to categorize only “pain”- and “free of pain”-related stimuli. However, the response mapping for the categories was reversed, assigning “pain” words to the “other” key and “free of pain” words to the “self” key. During this block each word was presented 6 times (i.e., 36 trials). In Blocks 6 and 7, participants again categorised as quickly and as accurately as possible the words of all 4 categories, using the other-pain response mapping. Pain words were now categorized by pressing the same key that was used for “other”-related words, whereas “free of pain” words were categorized by pressing the same key that was used for “self”-related words. Again each word was presented 4 times (i.e., 48 trials/block). Because we were primarily interested in interindividual differences in IAT effects, all participants completed the blocks in an identical sequence (see Table 1). The order of the presentation of the words within a block was completely random.

2.5. Procedure

Before the experimental session started (i.e., at the moment of scheduling the experiment session), patients with chronic pain (but not healthy volunteers) were invited to fill out a set of questionnaires at home (e.g., MPI, HADS, PDI, STAI-T, demographic information). They completed the questionnaires either online (via LimeSurvey) or on paper. Upon arrival all participants received general information about the study, signed an informed consent form, and filled out the STAI(-S). Thereafter, all participants performed several experimental tasks as part of the GPD-I study of which the IAT was one. During the IAT, on each trial a stimulus word appeared in the centre of the screen in white letters (Courier New, font size 14 pt) on a black background. Each word had to be assigned as quickly and as accurately as possible to one of the category labels (“me”, “not-me”, “pain”, “free of pain”) presented in the upper left and right corners of the screen by pressing the keys “A” for a left and “P” for a right response. If participants made a mistake, the message “wrong” appeared in red above the stimulus

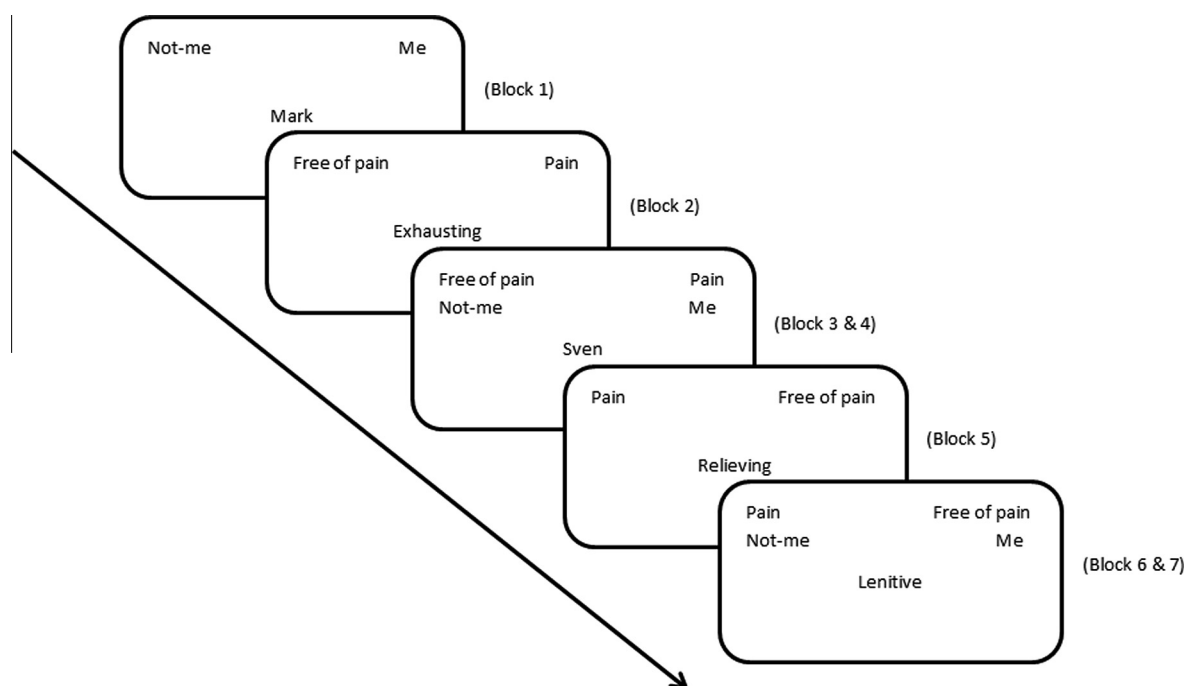


Fig. 1. Schematic representation of the Implicit Association Test (IAT).

Table 1

Trial types Implicit Association Test (IAT).

Block nr	Block name	Trials	Left key	Right key
1	Self - Other - discrimination	24	Other	Self
2	Pain - Free of pain discrimination	24	Free of pain	Pain
3	Self-pain practice block	48	Other + Free of pain	Self + pain
4	Self-pain test block	48	Other + Free of pain	Self + pain
5	Free of pain - Pain - discrimination	36	Pain	Free of pain
6	Other-pain practice block	48	Other + Pain	Self + Free of pain
7	Other-pain test block	48	Other + Pain	Self + Free of pain

for 400 ms. The inter-trial-interval varied randomly between 600 and 750 ms. Upon completion of the IAT, participants completed the ICQ and a manipulation check on which patients with chronic pain scored the “pain” and “free of pain”-related words on relevance for their particular pain on an 11-point Likert scale (0 = not at all relevant; 10 = very much relevant).

2.6. Data handling

In line with the present IAT literature, IAT scores were calculated by using the most often reported D600 scoring algorithm for IAT data, which has been shown to outperform conventional scoring algorithms [15]. When calculating the D600 index, which is the difference in RT between compatible and incompatible blocks, one has to include RTs on (mixed) practice blocks, add a 600 ms penalty (i.e., error penalty) to RTs on trials with incorrect responses, and correct the latencies for individual variability [15]. We implemented the algorithm in such a way that a positive D600 score reflects a stronger association between “self” and “pain” and “other” and “free of pain” than between “self” and “free of pain” and “other” and “pain” (i.e., a strong association between pain-schema and self-schema). A negative score reflects stronger associations between “self” and “free of pain” and “other” and “pain” than between “self” and “pain” and “other” and “free of pain” (i.e., less strong association between pain-schema and self-schema). Participants with error rates higher than 30% were treated as invalid and excluded from analyses. No participants needed to be excluded based on this criterion. As some individual difference variables (e.g., pain severity) did not have a normal distribution, we reported Spearman correlations when investigating the relationship with the D600 measure. Whenever possible effect size-indices for independent samples and the 95% confidence interval (95% CI) were calculated [4,6].

3. Results

3.1. Participant characteristics

The mean age of the participants was 49.35 years (SD = 10.01; range 21–65 years) and 83 were female (65.9%). Furthermore, a majority of the participants were married (52.0%) or living together (8.8%). Almost half of the participants graduated from high school or university (49.6%). In the patients with chronic pain, the median pain duration was 144.00 months (IQR = 161.00). Patients with chronic pain and healthy control subjects did not differ in terms of age or gender, but did differ significantly for education level and marital status (see Table 2 for an overview). Patients with chronic pain reported a mean disability level of 39.58 (SD = 11.32) on the PDI. Furthermore, they reported a mean pain level of 3.78 on the MPI (SD = 1.06). Almost all participants reported more than 1 pain location (M = 3.82, SD = 1.89; range = 1–9). Most commonly reported were back pain (90.4%), neck pain (67.1%), leg pain (65.8%), and arm pain (46.6%).

3.2. IAT

First, ratings of the manipulation check indicated that the allocation of words to the “pain” and “free of pain” category was appropriate. Patients with chronic pain reported that adjectives related to the pain category (M = 6.94, SD = 1.66) were more relevant for their pain compared with the adjectives that were allocated to the “free of pain” category (M = 3.09, SD = 2.31; $t(71) = 12.75$, $P < 0.001$, $d = 1.90$, 95% CI = 1.41:2.39). Next, we performed a one-way ANOVA with Group (chronic pain, healthy control) as between-subjects variable and D600 IAT score as dependent variable. Results indicated a main-effect of Group ($F(1, 124) = 30.16$, $P < 0.001$; $d = 0.99$, 95% CI = 0.62:1.37), indicating that D600 IAT score was larger in the chronic pain group (M = 0.12, SD = 0.51) than in the healthy control group (M = -0.33, SD = 0.35). To control for a possible baseline effect of education level and marital status analyses were repeated with both variables included as covariate. No influence of both variables was found (All $F_s < 1.55$) and the main effect of Group remained present ($F(1, 122) = 24.49$, $P < 0.001$).

3.3. Correlational analyses

For the patients with chronic pain, we calculated Spearman correlation coefficients between the D600 index and other individual difference measures (i.e., pain duration, state anxiety, trait anxiety, depression, disability, pain severity, pain suffering, helplessness, and acceptance; see Table 3). Results showed that pain severity (MPI-sev; $r = 0.23$, $P = 0.05$), pain suffering (MPI-suf; $r = 0.35$, $P < 0.01$), state anxiety (STAI-S; $r = 0.28$, $P < 0.05$), and trait anxiety (STAI-T; $r = 0.32$, $P < 0.01$) correlated significantly with the D600 index, indicating that patients with chronic pain with a stronger association between pain- and self-schema were also more anxious and reported more severe pain and suffering. Furthermore, we also found that the level of helplessness (ICQ-help; $r = 0.29$, $P < 0.05$) correlated significantly with the IAT index, indicating that a stronger association between pain and self-schema is related to increased feelings of helplessness.

4. Discussion

The main findings of the current study can be readily summarized. First, the pain- and self-schema were more strongly associated in patients with chronic pain than in healthy control subjects. Second, results indicate that, in patients with chronic pain, a stronger association between self- and pain-schema relates to a heightened level of pain severity, pain suffering, (state and trait) anxiety, and helplessness. In contrast with previous research, using explicit measures to assess the strength of the association between pain- and self-schema, no significant relationship was found between the strength of the association between pain- and self-schema and the level of acceptance or depression. Each of these findings deserves further attention.

Table 2
Participant characteristics.

	Patients with chronic pain (N = 73) [M (SD)]	Healthy control subjects (N = 53) [M (SD)]	t-value/ χ^2 -value
Age	49.95 (9.76)	48.53 (10.37)	0.78
Gender	65.8% Females	66.0% Females	0.001
Education level	1.4% Primary education; 59.2% secondary education; 39.4% higher education	3.8% Primary education; 32.7% secondary education; 63.5% higher education	8.61*
Marital status	61.6% Married; 8.2% cohabit; 21.9% alone; 6.8% divorced; 1.4% widowed	38.5% Married; 9.6% cohabit; 26.9% alone; 25.0% divorced	11.18*
STAI-S	38.01 (9.39)	29.92 (8.05)	5.06***
STAI-T	47.37 (11.09)	37.74 (10.49)	4.92***

STAI-S = State subscale of the State - Trait Anxiety Inventory; STAI-T = Trait subscale of the State - Trait Anxiety Inventory.

* $P < 0.05$;*** $P < 0.001$.**Table 3**
Means (M), standard deviations (SD) and Spearman correlation coefficients in patients with chronic pain (N = 73).

	M	SD	2	3	4	5	6	7	8	9	10	11
1. IAT index	0.12	0.51	0.14	0.23	0.35**	0.20	0.22	0.21	0.28*	0.32**	0.29*	0.02
2. Pain duration	169.77	111.83		-0.08	0.08	0.02	-0.07	-0.29*	-0.20	-0.14	-0.09	0.25*
3. Pain severity (MPI-sev)	3.78	1.06			0.46***	0.34*	0.21	0.31**	0.07	0.18	0.34**	-0.17
4. Pain suffering (MPI-suf)	4.15	1.21				0.52***	0.53***	0.36**	0.22	0.43***	0.37***	-0.39***
5. Disability (PDI)	39.58	11.32					0.36**	0.21	-0.10	0.29*	0.51***	-0.43***
6. Depression (HADS-D)	8.54	4.06						0.59***	0.42***	0.74***	0.50***	-0.54***
7. Anxiety (HADS-A)	8.67	3.87							0.46***	0.77***	0.40***	-0.42***
8. State anxiety (STAI-S)	38.01	9.39								0.62***	0.30*	-0.39***
9. Trait anxiety (STAI-T)	47.37	11.09									0.50***	-0.65***
10. Helplessness (ICQ-help)	15.56	3.95										-0.59***
11. Acceptance (ICQ-acc)	13.83	3.92										1

MPI = Multidimensional Pain Inventory; HADS-D = Depression scale of the Hospital Anxiety and Depression Scale; HADS-A = Anxiety scale of the Hospital Anxiety and Depression Scale; PDI = Pain Disability Index; STAI-S = State subscale of the State - Trait Anxiety Inventory; STAI-T = Trait subscale of the State - Trait Anxiety Inventory; ICQ = Illness Cognition Questionnaire.

ICQ, n = 72; HADS, n = 70.

* $P < 0.05$,** $P < 0.01$,*** $P < 0.001$.

First, results showed that the IAT index differed significantly between healthy control subjects and patients with chronic pain, indicating that pain- and self-schema are more strongly associated in patients with chronic pain than in healthy control subjects. This is in line with the idea that, although a schema (in this case the person's self-schema) within a person is relatively stable over time, repeated simultaneous presentation of elements from the pain- and self-schema may result in (partial) incorporation of one schema into another schema. The current finding is also in line with the schema-enmeshment model [33], which states that the self- and pain-schema become enmeshed to some degree in patients with chronic pain. Based on this idea, it might also be expected that for people who experience more severe pain, a simultaneous presentation of elements from pain- and self-schema will be more pronounced and thus result in a stronger association between these schemata. Results of the current study indeed point in this direction and indicate that a stronger association between pain- and self-schema is related to higher levels of pain severity (MPI) in patients with chronic pain (see also [25]).

Second, the strength of the association between pain- and self-schema was related to the level of suffering, anxiety, and the feeling of helplessness of the patients with chronic pain, but not to the level of acceptance. The fact that our IAT measure is related to higher levels of anxiety and suffering of the patients with chronic pain is in line with previous research using explicit measures (e.g., [42]) and assumptions of the schema-enmeshment model [33]. Indeed, the schema-enmeshment model suggests that stronger associations between pain- and self-schema could maintain and exacerbate anxiety/distress because the unique ability of pain

to interfere with cognitive functioning and interrupt most aspects of life [33]. Therefore, it could also be expected that the strength of association between pain- and self-schema would be related to the level of disability and suffering in patients with chronic pain. The current study indeed revealed that the strength of association between pain- and self-schema is positively related to patients' level of suffering, suggesting that indeed the enmeshment of pain- and self-schema relates to enlarged suffering in patients with chronic pain. The expected relationship with disability, however, failed to reach significance in this study.

Furthermore, in contrast with previous studies investigating the relationship between depression and the strength of association between self- and pain schema, results of this study failed to reach the conventional level of significance ($P = 0.07$). This inconsistent finding may partially be caused by the differences between the current sample and previous samples. The current sample was recruited via a self-help association, whereas the chronic pain populations in previous studies were mainly recruited in pain clinics [25,42]. For example, the level of depression of the current sample ($M = 8.54$, $SD = 4.06$) was significantly lower than in the sample of Sutherland and colleagues ($M = 9.99$, $SD = 4.38$; $t(154) = 2.138$, $P < 0.05$, $d = 0.34$, $95\% \text{ CI} = 0.03: 0.66$) [42]. Furthermore, in the current sample only 7 patients could be categorized as severely depressed (HADS-score ≥ 15 ; [39]), which may have reduced the chances to find a strong correlation between the strength of association between pain- and self-schemata and the presence of depression. To investigate the relationship with an individual difference measure of interest, future research might try to increase the variability in the study

sample or opt to compare pre-selected groups (high versus low) on the characteristic of interest. Also the absence of a relationship between the strength of the association between self- and pain-schema and acceptance is in contrast with previous findings of Morley et al. [25] who found that higher levels of self-pain-enmeshment were related to less acceptance (see also [42]). One reason may relate to the particular population included in each study. In line with this suggestion, Crombez and colleagues reported that the level of acceptance in a pain clinic sample was lower than in a self-help group [7]. A second reason that may explain the diverging results of the current study and previous research may relate to the different ways in which the self has been conceptualized. Indeed, the self is not a unitary construct. In the current study participants' "self" has been operationalized at its most fundamental level (i.e., "me" vs "not me"), whereas previous studies on self-pain-enmeshment used a more elaborated conceptualization of the self. These studies distinguished between attributes of the current self and attributes of future possible selves [42]. In doing so, the conceptualization of self-pain-enmeshment reflects the anticipation of non-goal attainment because of pain. It would be interesting to replicate our research using adaptations of the IAT that tap into other aspects of the self-concept (e.g., by using labels such as "I want to be" vs "I do not want to be" [8]). A last reason that might account for the discrepancy of results between the present study and previous studies investigating self-pain-enmeshment relates to the different paradigms that have been used to assess the strength of association between self- and pain-schema. It is possible that the IAT we used to investigate the relationship between self- and pain-schema does not measure the same construct as the (semi-)explicit measures which were assessed in previous research. Indeed, similar research in other domains has shown that dissociations can be found between explicit and implicit measures which assess a similar construct (e.g., [1,28,44]). An interesting direction for future research would be to combine the assessment of implicit measures and explicit measures of the strength of association between self- and pain-schema (1) to investigate whether both measures are related and (2) to investigate which measures are the best predictor for pain outcomes or changes caused by pain treatment [13,16]. Finally, this study indicates that a stronger association between pain- and self-schema is related to more feelings of helplessness. To our knowledge this study is the first study to investigate this relationship. Yet, the current finding is in line with the expectations as people are likely to perceive their pain as more uncontrollable when they perceive their pain to be stronger associated with their self-schema.

Present findings may have clinical and theoretical implications. First, our findings offer support for the schema-enmeshment model which suggests that the enmeshment of self- and pain-schema is present in patients with chronic pain and that this enmeshment is related to the level of pain severity and suffering [33]. Second, the IAT that we introduced offers an alternative way to assess the strength of the association between pain- and self-schema that is less susceptible to problems of response bias and is not restricted to conscious cognitive processes. Third, our research indicates that a stronger association between self- and pain-schema is related to negative pain outcomes and higher levels of anxiety and helplessness. This latter finding suggests that pain therapies may benefit from the inclusion of techniques that intervene on the level of self-pain-enmeshment.

Some aspects of the current study require further consideration. First, some correlational findings were of a small effect size ($0.25 < r < 0.30$). As our study is one of the first of its kind, replication of the current findings is necessary. Second, our research was cross-sectional and therefore cannot make any claim about the causal relationship between variables. To investigate the interest-

ing question of causality, future research might opt to use a longitudinal design. Third, our sample was recruited via an invitation letter. Only 10% actually responded which could have influenced the representativeness of the study sample. The characteristics of the current study sample are, however, comparable with samples of other studies (e.g., [29]). Fourth, in the current IAT, we opted to preselect pain-related stimuli that are commonly used in the Dutch language. Although a post hoc manipulation check confirms their relevance for this category, future research might allow people to select the most relevant terms for their particular pain. Fifth, one could argue that the IAT effect did not reflect associations between schema but merely the extent to which self and pain items are similar in terms of familiarity. Previous research on the IAT, however, demonstrated that overlap in terms of familiarity has little or no impact on IAT effects [31]. Last, although post hoc power analyses indicated that the current study had sufficient power ($\pi > 0.80$) to detect moderate effects ($r \geq 0.3$), small effects may have been missed because of a lack of statistical power.

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